

PROGRAMME 'APPLIED RESEARCH'

project application

ESFE 2025 CALL

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Authors names	Submission date	Group number

Project Duration: 4 months

Start date: October 2025

End date: February 2026

I. PROJECT

Project General Information
Title of the Project
<i>LumenCor – the educational heart model</i>
Project Abstract
<i>This proposal aims to develop an educational 3D heart model that visualizes cardiac blood flow, chamber occupancy, and the propagation of electrical impulses in real time, driven by electrocardiogram (ECG) signal analysis. The primary objective is to create a scientifically grounded, intuitive tool that directly links the heart's electrical activity to its mechanical function and anatomical pathways.</i> <i>The system will combine real-time ECG processing with a dynamic 3D representation of the heart. Key ECG features (P wave, QRS complex, T wave) will be automatically detected and mapped to the corresponding phases of the cardiac cycle. These signals will control both a physics-based simulation of chamber contraction, valve motion, and blood flow, and a set of strategically placed LED indicators embedded in the model. Each LED will illuminate according to the current location of the electrical impulse within the cardiac conduction system, providing a clear spatial representation of impulse propagation through the atria, atrioventricular node, and ventricles.</i> <i>Built on validated anatomical and physiological data, the model will allow users to observe in real time how electrical activation leads to coordinated mechanical action and hemodynamic changes. The integration of visual blood flow, chamber volume changes, and LED-based conduction mapping will offer a multi-modal learning experience that connects ECG interpretation with actual cardiac function.</i> <i>The platform is designed for educational use in medicine, biology, and biomedical engineering, as well as for patient education. In hospital settings, the model can be used to explain cardiac function to children and non-expert audiences in an accessible and engaging way. By bridging abstract ECG waveforms with tangible, observable heart dynamics, this project will provide an innovative teaching and communication tool that enhances understanding of cardiovascular physiology across diverse user groups.</i>
Project classification
Thematic area of the call
<i>Please indicate the thematic area of the call to which your proposal relates. Select one of the six options below:</i> - Welfare, health and care - <i>Digital and industry</i> - <i>Food and natural resources</i> - <i>Energy, transport and climate</i> - <i>Social and economic development</i> - <i>Unmanned vehicles</i> <i>If the proposal relates to more than one thematic areas please select the leading one.</i>
Key words
<i>ECG</i> <i>Model</i> <i>Heart</i> <i>Educational</i> <i>Pediatrics</i>

II. OBJECTIVES AND JUSTIFICATION

1. Current state of the art

The scientific state of the art relevant to this proposal combines three main strands: 3D cardiac anatomy visualization, mixed-reality and simulation-based teaching, and advanced cardiac electrophysiology and ECG analysis. Modern 3D printed and digital heart models are widely used to support anatomy and ECG interpretation, improving spatial understanding of structures, perfusion territories, and their relationship to surface recordings. Recent work in mixed and extended reality shows that immersive 3D presentations of cardiac anatomy linked to ECG waveforms can significantly enhance student engagement and understanding of the relationship between atrial/ventricular anatomy, electrophysiological properties, and P-wave or QRS morphology.

On the electrophysiology side, bidomain and monodomain computational models are considered the gold standard for simulating myocardial activation sequences and serve as the scientific basis for realistic electro-mechanical heart simulations, while parallel advances in deep learning and data-driven ECG analysis have delivered state-of-the-art methods for rhythm classification and disease prediction. Commercial educational platforms such as Epicardio already demonstrate real-time 3D visualization of cardiac electrical activity synchronized with simulated ECG, confirming both the technical feasibility and pedagogical value of linking waveforms with spatial conduction maps. However, existing tools typically focus either on electrical activation or on anatomy and blood flow, and they usually rely on internally generated signals rather than mapping arbitrary external ECGs to real-time chamber occupancy.

This project builds on these developments by introducing an educational 3D heart model that couples real-time ECG feature detection (P wave, QRS complex, T wave) with a tangible, physics-based representation of chamber contraction, valve motion, blood flow and LED-based conduction pathways. In earlier work, the market landscape of heart-education tools was analyzed, revealing gaps in real-time ECG-driven chamber filling visualization and opportunities for low-risk, education-focused devices that remain physiologically grounded without providing clinical decision support; this proposal directly targets those gaps.

2. Project objectives

The overall objective of this project is to design, build and test an educational 3D heart model that links real-time ECG signals with a tangible visualization of cardiac anatomy, blood flow and electrical conduction, in a way that is scientifically sound and easy to understand for non-experts.

Specific objectives are:

- 1. Define and implement an ECG-processing module that detects key features (P wave, QRS complex, T wave) and maps them to phases of the cardiac cycle (atrium filling/emptying, ventricular systole and diastole) in real time.*
- 2. Develop a 3D heart model (digital and/or physical) with physics-based animations of chamber contraction, valve motion and blood flow, driven by the ECG derived timing signals.*
- 3. Integrate a set of LED indicators representing the main elements of the conduction system (SA node, AV node, His–Purkinje network) and synchronize their illumination with the detected propagation of the electrical impulse.*
- 4. Validate the model's anatomical and physiological plausibility with domain experts and refine it based on structured feedback (e.g. expert review sessions, small user tests with students).*

3. Methods, approach and concept

The project will use an iterative prototyping and evaluation approach over one year, combining hardware development, embedded signal processing and user testing in a small team. The core method is to build a hybrid system consisting of a physical 3D heart model with strategically placed LEDs (one in each atrium and ventricle) and a companion mobile application that displays the live ECG trace. Real-time ECG data will be acquired from a volunteer using an AD8232-based heart monitor connected to an ESP-WROOM-32 microcontroller, powered from a portable power bank, and streamed simultaneously to the LEDs and to a smartphone for plotting.

The underlying concept is to map detected ECG features (P wave, QRS complex, T wave) to simplified phases of the cardiac cycle, and to express these phases through visible changes in chamber “activation” (LEDs) and the live waveform, rather than through full biophysical fluid or finite-element models. This relies on established electrophysiological theory linking surface ECG components to atrial and ventricular depolarization and repolarization, but implements it in an intentionally didactic, low-complexity form suitable for teaching. The main assumptions are that simplified, phase-based mappings are sufficient for educational purposes, and that real-time, beat-to-beat responses are more engaging and intuitive than pre-rendered animations.

To reach the objectives, work will proceed through: (1) defining ECG feature-detection and phase-mapping algorithms on the ESP32, (2) designing and fabricating the 3D heart model and LED layout, (3) developing the mobile app for live plotting and interaction, (4) integrating and debugging the full system, and (5) running a small user study with students alongside expert reviews by professors in relevant disciplines. The project is positioned as an educational, non-diagnostic tool at approximately Technology Readiness Level 4: a lab-tested prototype demonstrated in a relevant educational environment, with clear potential for further refinement but not yet a market-ready product.

4. Market need identification

Market demand for project results

There is a growing demand in medical, nursing and biomedical-engineering education for tools that connect ECG interpretation with 3D visualizations of cardiac function, rather than teaching them separately. Existing solutions either show generic blood flow in 3D apps or simulated ECG in manikins, but very few products provide an inexpensive, classroom-ready system that responds in real time to a learner’s own ECG and makes chamber filling and conduction pathways visible on a tangible model. This creates clear demand from universities, skills labs and hospitals for an engaging, low-cost device that can be used repeatedly in teaching and patient explanation.

Characteristics of the target market

The primary target market consists of universities and colleges running medicine, nursing, biology and biomedical-engineering programs, particularly courses on cardiovascular physiology, pathophysiology and ECG interpretation. A secondary market includes hospital education departments and simulation centers that need intuitive tools to explain heart function to students, junior staff and patients, especially children and non-experts. These users value low maintenance, portability, clear didactic value and the ability to integrate the device into existing teaching sessions without complex setup.

Competition

Competing solutions include 3D anatomy and AR/VR platforms such as Visible Body, BioDigital Human and HeartWorks, which offer rich cardiac visuals and schematic blood flow but rely on pre-defined animations instead of live ECG input. High-fidelity patient simulators from Laerdal and Gaumard integrate ECG and other vitals into realistic scenarios, yet they simulate ECG internally and are expensive, lab-scale systems rather than simple, dedicated heart-education tools. There are also

specialized platforms like Epicardio and various research prototypes that link electrophysiology to 3D activation, but these are niche, software-centric and not typically available as low-cost, physical classroom devices.

5. Novelty of the project results

Description of the novelty of the project results

The project is novel in that it combines a physical 3D heart model, live ECG acquisition and a mobile visualization into one affordable, education-focused system that reacts in real time to a volunteer's own heartbeat. Unlike existing anatomy apps or simulators, which either play pre-rendered cycles or generate synthetic ECG internally, this prototype uses an external ECG board and embedded processing to drive both chamber-specific LEDs and a live plot, directly linking waveform features to spatial and mechanical changes. It therefore advances beyond current state-of-the-art products by offering a concrete, hands-on way to experience ECG–mechanics coupling without aiming at clinical decision support.

New features and functionalities of the project results

New functionality of the project results

Feature / functionality of the project result.	Benefit / advantage	Technical parameters
Real-time mapping of an externally measured ECG onto chamber-specific LEDs in a physical 3D heart model.	Makes the link between ECG waveform and activity of each atrium and ventricle immediately visible and intuitive for learners.	AD8232 ECG module + ESP-WROOM-32 microcontroller, 4 LEDs, sampling and update in real time
Simultaneous live ECG plotting on a smartphone application connected to the same hardware.	Allows students to see the waveform and the heart model react together, supporting ECG reading skills and understanding of conduction timing.	Wireless data transfer from ESP-WROOM-32 to mobile app, continuous ECG stream from AD8232, real-time plotting on phone screen.
Portable, low-cost hybrid device dedicated to education	Affordable and easy to deploy in ordinary classrooms and skills labs, unlike large, expensive high-fidelity simulators.	Battery-powered via power bank, compact 3D-printed heart model, off-the-shelf ESP-WROOM-32 and AD8232 hardware, TRL 4 prototype.

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Intellectual Property Rights (IPR) management

The management of Intellectual Property Rights within the consortium will be formally governed by a Partnership Agreement, which will be signed and delivered prior to the conclusion of the project contract with the Program Operator. The project is expected to generate distinct categories of IP that require specific protection strategies to ensure future commercial exploitation. The firmware and software code developed for the ESP32 and mobile application will be protected under copyright, while the specific industrial designs of the 3D heart model—particularly the internal conduit system for LED integration—will be managed as proprietary trade secrets or industrial designs. Furthermore, the business strategy, including the "Over-the-Air" (OTA) update mechanism and specific pathology software packages, will be treated as confidential know-how to secure the project's scalable revenue model.

Intended short-term outcomes

In the short term, the primary outcome of the LumenCor project will be the delivery of a fully functional educational tool at Technology Readiness Level 4, defined as a lab-tested prototype demonstrated in a relevant educational environment. By the end of the four-month period, the team will have successfully integrated the electronic hardware with the 3D-printed anatomical model, resulting in a device that visualizes real-time ECG signals via synchronized LED indicators. Parallel to hardware development, a stable mobile application will be finalized to allow for simultaneous live ECG plotting via Bluetooth with low latency. From a regulatory and business perspective, short-term results will include a comprehensive Electrical Safety Report verifying compliance with IEC standards and a finalized pitch deck backed by a comparative market analysis. These activities will also foster significant acquisition of expertise for the project team in the fields of biomedical signal processing, embedded systems engineering, and additive manufacturing.

Intended long-term application of outcomes

In the medium to long term, the project intends to bridge the "Abstract to Concrete Gap" in medical education by transitioning the prototype into a mass-market commercial product. The strategy

involves deploying the device to universities, nursing colleges, and biomedical engineering programs to serve as an affordable alternative to high-fidelity simulators, utilizing a low manufacturing cost to disrupt the current market pricing structure. A key long-term application includes the expansion into hospital settings as a "Certified Pediatric Distraction Buddy," where the device will be used to explain cardiac conditions to children and non-expert patients in an accessible, engaging manner. The project aims to establish a sustainable economic impact through a scalable revenue model based on wireless software expansion, where users can purchase specialized pathology software packages to simulate arrhythmias on the existing hardware base.

Ethical issues

Ethical considerations for the LumenCor project have been assessed using the Ethics Issues Table Checklist and center primarily on human safety and data privacy during the use of the device. Because the prototype involves connecting electrodes to human participants to acquire real-time ECG signals, there is a potential electrical risk. This will be managed by strictly adhering to IEC 60601 standards for medical electrical equipment, ensuring galvanic isolation through battery operation, and implementing hardware current limitations. To avoid the regulatory and ethical burdens of clinical trials, the system is explicitly classified as an educational teaching aid rather than a clinical medical device, and legal disclaimers will clearly state that it is not for diagnostic purposes. Furthermore, the project will enforce data privacy protocols and require Informed Consent Forms for all participants to ensure they understand how their physiological data is processed and visualized.

III. WORK PACKAGE DESCRIPTION

1. Work Package Description

In section 'III. WORK PACKAGE DESCRIPTION' of an application form each work package (max. 5) should be described. For each work package (WP) a separate sheet should be added. Please present work packages in detail. The explanations have to be sufficient for justifying the proposed effort and allow for progress monitoring.

PROJECT MANAGEMENT AND COORDINATION

WP No.	1
WP leader (name/title)	Agata Kraska, Team Leader
WP Title	Project management and coordination
WP duration	4
Start Month	1
End Month	4
WP objectives	
To ensure the efficient implementation of the LumenCor project within the scheduled time. The objective is to coordinate the technical, design and legal teams, manage resources and monitor progress to mitigate risks and delays.	
Description of work	

Task 1.1: Coordination and Communication <i>Organization of weekly internal meetings to track progress. Allocation of tasks between the Engineering (WP2), Design (WP3), and Legal (WP4) teams to ensure smooth workflow.</i>	
Task 1.2: Financial and Resource Management <i>Management of the budget and procurement of components (ESP32, Sensors, Filament). Tracking the "Bill of Materials" (BOM) to keep the cost per unit under the target.</i>	
Task 1.3: Risk Management <i>Identification of potential safety risks and ensure of compliance with relevant EU regulations.</i>	
Deliverables	
D1.1 - Project Plan (Month 1): Timeline and definition of roles.	
D1.2 - Midterm presentation (Month 2): Presentation summarizing the preliminary results of the project.	
D1.3 - Final Cost Report (Month 4): Final Bill of Materials (BOM) and economic analysis.	
D1.4 - Final Presentation (Month 4): The complete pitch deck summarizing the project results.	
Milestone	
Description of milestone	M1 - Project Kick-off (Month 1): Team assembled and roles assigned.
	M1.2 - Project Completion (Month 4): All WP outputs delivered and ready for presentation.

HARDWARE AND SOFTWARE ENGINEERING AND SIGNAL PROCESSING IMPLEMENTATION	
WP No.	2
WP leader (name/title)	Alessandro Zambon, lead developer
WP Title	Hardware/Software Engineering and Signal Processing Implementation
WP duration	4
Start Month	1
End Month	4
WP objectives	

The objective of this WP is to translate the theoretical models defined in WP1 into a robust, physical electronic system. We aim to engineer a stable IoT device capable of acquiring high-fidelity ECG signals and transmitting them simultaneously to a physical actuator (3D heart model) and a digital interface (smartphone).

Specific Objectives:

1. **Circuit Engineering:** To design and assemble a noise-resistant circuit integrating the AD8232 analog front-end with the ESP32-WROOM.

2. **Firmware Deployment:** To develop optimized MicroPython firmware handling ADC reading, signal processing, PWM LED driving, and BLE communication concurrently.

3. **Power Management:** To ensure the system operates stably via external power as required by safety regulations.

Description of work

Task 2.1: Electronic Circuit Design & Assembly

Selection and wiring of components. We will interface the AD8232 sensor via the ADC pins of the ESP32, implementing hardware decoupling capacitors to reduce power supply noise.

Task 2.2: Embedded Firmware Development

Writing the production code in MicroPython. The firmware structure will be multi-threaded or interrupt-based to ensure that the heavy mathematical processing (Squared Energy calculation) does not block the Bluetooth transmission or the smooth PWM dimming of the LEDs.

Task 2.3: Wireless Connectivity (BLE)

Development of the communication protocol to stream raw ECG data and "Peak Detected" flags to a mobile device with low latency (<50ms).

Deliverables

D2.1 - Circuit Prototype (Month 1): The fully assembled electronic stack (ESP32 + AD8232 + Wiring).

D2.2 - Firmware V1.0 (Month 3): The complete, bug-free code uploaded to the microcontroller.

D2.3 – Complete circuit (Month 4): Complete and soldered circuit, ready for implementation.

D2.4 - Connectivity Module (Month 4): Verification of stable Bluetooth transmission to a smartphone app.

Milestone

Description of milestone

M2 – First prototype (Month 2): The electronics are fully assembled and the ESP32 successfully reads a signal from the sensor.

M2.1 – Final prototype (Month 4): The circuit is fully soldered together, the ESP32 can read and send the signal to the 3D model and to the telephone.

3D DESIGN, PROTOTYPING & SYSTEM INTEGRATION	
WP No.	3
WP leader (name/title)	Mateusz Bernacki, design lead
WP Title	3D Anatomical Modelling and Physical Integration
WP duration	4
Start Month	1
End Month	4
WP objectives The objective of this WP is to bridge the gap between abstract ECG signals and physical physiology by developing an anatomically accurate 3D heart model through additive manufacturing. This model will feature hollow internal cavities representing the atria and ventricles, specifically designed with integrated conduits to house the electronic wiring and LED arrays required for real-time visualization. By utilizing translucent materials, the model will allow internal light pulses to diffuse through the structure, simulating blood flow and chamber occupancy synchronized to the user's heartbeat. The system focuses on creating a multi-modal learning experience that links heart's electrical activity to its mechanical function through tactile, observable interface.	
Description of work Task 3.1: 3D Modelling and Anatomical Design Development of a multi-part heart assembly featuring hollow cavities for all four cardiac chambers. The design will incorporate integrated internal chambers and conduits to route wiring from the sensors to the LEDs without obstructing the anatomical representation. Task 3.2: Material Optimization and Additive Manufacturing Selection of biocompatible translucent PETG filament to ensure structural durability and optimal light diffusion. The manufacturing process will involve iterative testing of wall thickness to balance the model's opacity with the visibility of internal LED pulses. Task 3.3: Mechanical Assembly and LED Integration Manual installation of strategically placed coloured LEDs within the heart structure to symbolize physiological states. This stage includes the final wiring and structural connection of the heart model to the hardware base containing the ESP32 and AD8232 stack.	
Deliverables D3.1 – Final design files (Month 2): Finalized files of the heart model featuring internal conduits and the electronic housing base D3.2 – Printed Anatomical Model (Month 3): A physical heart prototype manufactured using translucent PETG to ensure optimal light diffusion D3.3 – Integrated and Functional LumenCor Prototype (Month 4): The complete assembly of the 3D model, internal LED arrays and the hardware stack, resulting in a fully operational educational tool	
Milestone Description of milestone	
M3 – System Integration Completion (Month 4): Final assembly of the 3D-printed heart and electronic stack completed, with verified LED synchronization	

MARKET ANALYSIS & BUSINESS STRATEGY	
WP No.	4
WP leader (name/title)	Anna Benfatto, business strategy
WP Title	Market Gap Analysis and Commercial Strategy
WP duration	4
Start Month	1
End Month	4
WP objectives The objective of this WP is to define LumenCor's market positioning as a disruptive educational tool by addressing existing market gaps in medical simulation. This includes identifying primary targets across medical universities and paediatric healthcare to bridge "Simulator Price Gap" and "Abstract to Concrete Gap". The strategy focuses on demonstrating a high-fidelity experience at a fraction of the cost of clinical simulators. The system aims to establish a scalable revenue model based on low manufacturing costs and wireless software expansion.	
Description of work Task 4.1: Comparative Market Analysis Performance of a detailed analysis against competitors like EPICARD and Visible Body to highlight LumenCor's price advantage. This task quantifies the "Disruptive Economics" by comparing the 150zł manufacturing cost against the 180 000zł+ cost of high-end simulators. Task 4.2: Business Model and Target Definition Definition of the primary business model focusing on mass-market educational kits for universities. Research includes expansion strategies into the "Certified Paediatric Distraction Buddy" market for hospital settings. Task 4.3: Scalable Revenue and OTA Strategy Development of an Over-the-Air (OTA) update strategy to increase product lifespan and utility. This includes planning for the sale of specialized pathology software packages that utilize the existing hardware base without requiring new physical components.	
Deliverables D4.1 – Market and Competitor Analysis (Month 2): Comprehensive report identifying market gaps and performance comparison with existing clinical tools D4.2 – Financial Plan and Pricing Strategy (Month 4): Final economic analysis including Bill of Materials (BOM), suggested MSRP (750zł – 1200zł) and target markups	
Milestone Description of milestone	
	M4 – Market Strategy Finalization (Month 4): Completion of the commercial roadmap and pitch deck ready for educational and clinical dissemination

REGULATORY COMPLIANCE, SAFETY & TESTING	
WP No.	5
WP leader (name/title)	Agata Kraska
WP Title	Safety regulations and Educational Certification
WP duration	4
Start Month	1
End Month	4
WP objectives	
The objective of this WP is to guarantee that the LumenCor device is safe for human use and complies with the educational standards. This includes defining the medical classification to avoid unnecessary clinical regulatory burdens while rigorously managing electrical risk associated with human electrode contacts. The system is designed to operate as an “educational/teaching aid” rather than a clinical medical device.	
Description of work	
Task 5.1: Compliance and Electrical Safety Analysis Performance of safety audit based on IEC 60601 standards for medical electrical equipment. Work includes documenting safety choices such as current limitation, the use of insulated power sources for galvanic isolation and ensuring low-voltage operation. Task 5.2: Implementation of Safety Monitoring system Development of an electrode detection system using the LO+ and LO- digital output pins of the Ad8232 sensor. This task involves programming the ESP32 to monitor connection status and alert the user if electrodes are properly attached to the body or disconnected. Task 5.3: Legal Documentation and Ethical Compliance Drafting of essential legal disclaimers stating the device is “not for diagnostic purposes” to reduce regulatory requirements. This task also includes creating informed consent forms for participants and establishing data privacy protocols for all collected ECG signals.	
Deliverables	
D5.1 – Electrical Safety Report (Month 3): Documentation of safety engineering choices, including current limitations and battery isolation verified against IEC 061-1-2 standards.	
Milestone	
Description of milestone	M5 – Internal Safety Certification (Month 4): Completion of a proper ethical screening and internal safety verification ensuring the prototype is ready for educational deployment.